Effects of Air-to-Fuel Ratio on Gas Turbocharged Engine Performance

Word Count: 4973

Abstract

Given increasingly stringent emissions regulations and increased global warming, there is a global need for optimized engine performance. Researchers and car companies are turning towards turbocharged engines to address this need. This study focuses on the effects of Air-to-Fuel Ratio (AFR) on the performance of a gas turbocharged engine, where performance is measured through engine knock, torque output, and brake-specific fuel consumption. A simplified mean value engine model was created in order to simulate the fundamental internal thermodynamic behaviors of the engine under isentropic and steady-state thermal conditions under a wide range of engine speed conditions. The data was then processed using a min-max normalization in order to determine the most optimal AFR for the following conditions: the highest output torque, the lowest engine knock, the lowest fuel consumption, and the lean best torque. Findings show that the best λ for maximizing torque output and minimizing engine knock is 0.8. The best λ for fuel efficiency is 1.2. The λ for the lean best torque output was 0.9. Results indicate that leaner conditions are better for engine efficiency and that richer conditions are better for increasing torque and reducing engine knock. Additionally, a slightly rich mixture is best for balancing torque output, fuel consumption, and engine knock. These results align with the general trends observed in the existing literature, and it can be concluded that AFR trends for naturally aspirated gas engines generally align with those seen in gas turbocharged engines.

Keywords: air-to-fuel ratio, gas turbocharged engine, mean value engine model, brake-specific fuel consumption, torque output, engine knock

Nomenclature

Table 1

Variables	**Symbol**	**Units**
Air to Fuel Ratio (AFR)	AFR	N/A
Brake Power	BP	kilowatts
Brake Specific Fuel Consumption	$BSFC$	$\frac{kilograms}{kilojoules}$
Engine Speed BTE Contribution	η_{thn}	N/A
Ignition Delay Time	T_{ID}	milliseconds
Intercooler Temperature	T_{ic}	Kelvin
Knock Margin	KM	milliseconds
Manifold Absolute Pressure	p_{man}	Pascals
Manifold Pressure BTE Contribution	$\eta_{thp_{man}}$	N/A
Mass Air Flow	m_{air}	$\frac{kilograms}{second}$
Mass Fuel Flow	m_{fuel}	$\frac{kilograms}{second}$
Maximum Brake Torque Spark Angle	θ_{mbt}	Degrees
Normalized AFR (used synonymously with AFR in this paper)	λ	N/A
Normalized AFR BTE Contribution	$\eta_{th\lambda}$	N/A
Normalized BSFC Score	Z_{BSFC}	N/A
Normalized Knock Score	Z_{knock}	N/A
Normalized Performance Score	Z_{P}	N/A
Normalized Torque Score	Z_{T}	N/A
Peak Pressure	p_{peak}	Pascals
Peak Temperature	T_{peak}	Kelvin
Pressure Ratio	p_{r}	N/A

Residence Time	T_{RES}	milliseconds
Spark Angle	θ	Degrees
Spark Angle BTE Contribution	$\eta_{th\theta}$	N/A
Spark Angle Correction Term	n_{47}	Degrees
Temperature After Compressor	T_{c}	Kelvin
Temperature After Intercooler	T_{cool}	Kelvin
Temperature After Throttle	T_{man}	Kelvin
End Gas Temperature	T_{EGT}	Kelvin
Torque	τ	Newtons * Meters
Total Brake Thermal Efficiency (BTE)	η_{th}	N/A
Volumetric Efficiency (VE)	η_{vol}	N/A

Table 2

Constants	**Symbol**	**Values**	**Units**
$Ambient\ Pressure$	p_{amb}	101,700	Pascals
$Ambient\ Temperature$	T_{amb}	296	Kelvin
$Compression\ Ratio$	CR	8.71	N/A
$Gas\ Constant$	R	287	$\frac{Joules}{(kilograms)(Kelvin)}$
$Gas\ Octane\ Number$	ON	92	N/A
$Heat\ Capacity\ Ratio$	γ	1.4 (initially)	N/A
$Heating\ Value\ of\ Fuel$	Q_{heat}	$44,000$	$\frac{kilojoules}{kilogram}$
$Volumetric\ Displacement$	V_d	$2.5 * 10^{-3}$	Meters cubed

Parameters	**Symbol**	**Values**	**Units**
$Boost\ Pressure$	p_{boost}	48,000 — 110,000	Pascals
$Converted\ Engine\ Speed$	n	2.75 — 5.5	1000 x Revolutions per Minute
$Engine\ Speed$	N	2750 — 5500	Revolutions per Minute
$Intercooler\ Efficiency$	ϵ	0.76 — 0.87	N/A

* Parameters were tested at discrete, non-linear intervals

Background

Note: Throughout this paper, the term AFR is used synonymously with lambda (λ) to represent the normalized air-fuel ratio.

Introduction

Internal Combustion Engines (ICEs) are one of the most common types of engines powering vehicles today. They produce power through the combustion of an air-fuel mixture that produces exhaust gases as a byproduct [1, 2, 3]. Given the polluting nature and popularity of ICEs and other processes, greenhouse gases and emissions have spiked, and emissions regulations have become more stringent in recent years [4]. These concerns have led the European Union, Environmental Protection Agency (EPA), and the Department of Transport to impose limits on emissions to limit pollution [1, 5]. International regulations like these have driven engine development toward fuel efficiency and lower pollution to address this global challenge [6, 7]. As a result, researchers are looking for engine optimization methods to minimize pollution and global warming [8].

Currently, researchers are increasingly focusing on downsized Gas Turbocharged Engines (GTEs) to address these environmental concerns [9] because it is known that turbochargers and engine downsizing enhance ICE fuel efficiency and reduce emissions while maintaining solid performance that is comparable to larger nonturbocharged engines. [1, 10, 11]. Consequently, this technology is very attractive to many car manufacturers. [9, 12]. One example is Ford's EcoBoost program, which has successfully engineered downsized GTEs to meet these new government regulations for fuel and emissions [13]. In addition to GTEs, the Air-to-Fuel Ratio (AFR), which is the mass ratio of the air and fuel contents within the combustion mixture [10], plays a massive role in the torque output and fuel efficiency of the engine. This makes AFR vital for addressing the global need for more efficient and environmentally friendly engines. Combining AFR optimization methods with downsized GTEs allows for the maximization of engine efficiency and performance.

Fundamental Concepts

AFR is categorized as rich, lean, or stoichiometric. Lean fuel mixtures that have more air mass than fuel mass and have a higher AFR; rich mixtures have more fuel mass and less air mass and have a lower AFR [14]. It is commonly represented by lambda (λ); at $\lambda = 1$, the AFR = 14.7; at leaner mixtures, $\lambda > 1$; for richer mixtures, $\lambda < 1$ [15]. It is implied that the stoichiometric ratio is the point where all the fuel mass reacts with all the air mass [15]. Generally, lean conditions produce hotter temperatures (increasing engine damage risk), and richer conditions produce lower temperatures (reducing engine damage risk) [15].

Performance reflects the overall functionality of an engine [16]. Torque, the rotational force produced by the air-fuel mixture [17, 18], is a key metric in determining engine power [16]. Power is used to measure engine performance, and power is directly proportional to torque; thus, higher torque means a stronger engine performance [7, 16]. Torque in this paper will be measured under Wide Open Throttle Conditions (WOTC), which refers to the throttle position in which the maximum amount of air and fuel enter the engine, resulting in maximum torque output [19]. This throttle condition was used due to its wide applicability and standardizability across different engine types.

"Engine speed is defined as a key parameter that affects certain engine performance characteristics, and it's typically measured in Revolutions Per Minute (RPM)" [20]. The RPM range for this study is from 2750 RPM to 5500 RPM. However, the main focus of this study is on AFR. The RPM range is included to encompass various driving conditions for research practicality.

Brake Specific Fuel Consumption (BSFC) is a metric measuring the amount of fuel an engine uses relative to the power it produces; it is a commonly used metric for engine efficiency, and thus it was selected for this experiment [17]. A higher BSFC value results in lower fuel efficiency because the engine uses more fuel to generate the same amount of power; conversely, a lower BSFC value means greater fuel efficiency.

Knock, the preignition of fuel mixture before spark in ICEs, can lead to engine damage, increased fuel consumption, and decreased engine performance and efficiency [12, 3, 21]. Furthermore, knock occurs more frequently at higher engine temperatures [1, 10]. Engine knock will be utilized as an engine operation safety metric in this study. A higher knock probability indicates a greater risk of engine damage.

Literature Review

AFR and BSFC/Torque

Khan's experiment on Air-to-Fuel Ratio (AFR) and the effect of various ethanol fuel blends has concluded that richer AFR produces more engine torque across all engine speeds [8]. In a similar ethanol blend study by Wu et al., a slightly rich AFR results in maximum torque and minimal fuel efficiency [22]. Building on this idea, Andersson's experiment exploring cylinder air charge estimation for optimal AFR control strategies in Gas Turbocharged Engines (GTEs) shows that high torque conditions require rich AFR values to lower engine temperatures under various Revolutions per Minute (RPM) conditions [10].

In contrast, leaner conditions result in more efficiency. Frasci et al. explore the effects of leaner AFR in their study on engine pre-chamber design and AFR's effects on gas engine fuel consumption. They effectively conclude that leaner AFR results in more fuel efficiency under higher RPM conditions [5]. The study by Wu et al. also confirms these findings for additional RPM ranges [22]. Furthermore, in the Nemati et al. 2024 study, which focused on comparing different controllers (SMC and PID controllers) for their effectiveness on AFR control and their ability to reduce emissions, straying too far from the stoichiometric ratio results in reduced fuel efficiency and torque [2]. The previous ideas are summarized by the Environmental Protection Agency (EPA) facts sheet; leaner conditions promote fuel efficiency but reduce engine power, and the converse is true for richer conditions [23]. Based on the preceding studies, it can be assumed that richer conditions produce more power with less fuel efficiency, while leaner conditions produce the inverse effect.

AFR and Knock

A study by Almaleki et al. explored the effects of different fuel blends and their impact on engine knock at low RPM ranges, ultimately concluding that AFR has a massive impact on engine knock occurrence; however, its focus was primarily on AFR being a supplemental variable to the combustion ratio [21]. Frasci et al. expand on this claim, stating that leaner AFR has a higher chance for knock [5]. This is because, according to Speight, who explores the various properties of different fuel compositions, leaner conditions result in a higher engine temperature. [15]. On the richer side, Andersson states that rich mixtures cool engine temperatures, which reduces engine knock within a GTE context [10]. Finally, in a study by Amann et al. studying Low-Speed Pre-Ignition (LSPI), it’s established that LSPI is an abnormal combustion phenomenon directly correlated with knock in gas turbo engines, and that leaner AFR can increase LSPI and that richer AFR can reduce LSPI under low-speed conditions. However, it’s important to note that this study was largely focused on the impact of fuel blends and aromatics on knock rather than solely focusing on AFR [9]. It's evident that engine temperature is directly correlated with knock. Given that AFR is responsible for engine temperature, AFR is a key variable influencing knock.

AFR and Turbocharged Engines

It’s important to note that the following definitions and relationships were explored with a primary focus on gas naturally aspirated engines. The studies by Amann et al. [9] and Andersson [10] were the only studies focused on the relationship between AFR, knock, and torque in a turbocharged gas engine context. Their work for GTEs aligns with the general trends of AFR, knock, and torque for naturally aspirated engines. However, they did not encompass Brake-Specific Fuel Consumption (BSFC) in their findings.

Gap

Existing research is primarily focused on AFR control and AFR's impacts on performance under different fuel blends. There have been few studies on AFR's direct effects on a turbocharged gas engine. Moreover, a majority of studies used AFR as a secondary variable to draw conclusions about other engine variables rather than being the sole focus of the study. Additionally, most of the studies focused on knock, performance, and fuel efficiency separately; they never included the entire breadth of the variables within a single study. Finally, there has been no study found that creates a custom calculator that calculates an engine optimization score for efficiency, knock, and torque for a GTE, or a study that fully models the torque, knock, or BSFC values without the use of a physical engine. Addressing these gaps, this study will examine the direct effects of AFR conditions on torque, knock, and fuel efficiency within a turbocharged application over a wide range of steady-state RPM conditions to find an optimal balance for engine torque, knock, and BSFC. This paper addresses the following research question: How do rich, lean, and stoichiometric AFR values impact torque output, BSFC, and knock tendency at Wide-Open Throttle Conditions (WOTC) in GTEs, and what optimal AFR ranges maximize performance and efficiency while minimizing engine knock?

This research hypothesized that for GTEs under WOTC, a richer AFR would result in reduced knock and higher torque compared to leaner mixtures. This hypothesis aligns with the GTE studies by Amann et al. [9] and Andersson [10], as well as the other gas engine studies showing that richer AFR lowers combustion temperatures to reduce engine pre-ignition and create optimal operating temperatures to produce maximum power. Going with the general gas engine study trends, BSFC is expected to increase (worse fuel efficiency) for richer mixtures and decrease for leaner mixtures (better fuel efficiency). The highest torque and lowest knock conditions are expected to occur at slightly rich AFR ($\lambda \approx 0.90$-0.95), while the most fuel-efficient conditions are expected to occur at slightly lean AFR ($\lambda \approx 1.05$-1.1). Through this data, it's predicted that the most optimal AFR value for the highest torque output that minimizes fuel consumption and engine knock tendency (lean best torque) will be slightly rich ($\lambda \approx$ 0.95-1.00) in order to preserve fuel efficiency and limit engine knock without sacrificing too much torque output. These conclusions align with the studies showing that optimal power and efficiency values cluster near the stoichiometric ratio.

Methodology

Model Selection

This study aims to evaluate the impacts of various Air-to-Fuel Ratio (AFR) conditions on Brake-Specific Fuel Consumption (BSFC), torque, and knock over a wide range of Revolutions Per Minute (RPM) conditions under Wide-Open Throttle Conditions (WOTC) for a Gas Turbocharged Engine (GTE) in order to determine the optimal AFR range for GTE operation. The goals are achieved through the creation of a Mean Value Engine Model (MVEM) that is theoretical in nature. An MVEM model is a computational function that models the various engine cycles while simplifying certain engine dynamics [10]. A majority of scholarly studies used a real engine or real-time simulation [2, 4, 8] to gather engine performance data, but to successfully explore the effects of an engine beyond the typical limitations of AFR, an MVEM model was selected for this study. Additionally, the cost-effectiveness, speed compared to real-time simulation, superior variable control, and its relative accuracy and precision [10, 24, 25] compared to a real engine further informed this decision. Some studies independently modeled each subsystem before coupling them into a comprehensive MVEM, while others validated MVEMs through the use of a real engine. This study was unable to implement the preceding procedures due to resource limitations and limited coding experiences. Given the lack of access to an engine and limited access to literature detailing specific software implementation strategies, a simplified mathematical approach was used. This approach utilized equations that were validated by other papers through real engine testing. A 0-dimensional, mathematical, semi-empirical, quasi-steady-state, input-output model was created in order to simulate the fundamental internal thermodynamic behaviors of the GTE under steady-state thermal conditions [25]. This means that the system is ideal, models reversible processes, and is highly simplified when compared to professional models. The inputs were RPM and desired AFR, and the outputs were BSFC, torque, and knock.

Mathematical Formulations

AFR/RPM Condition

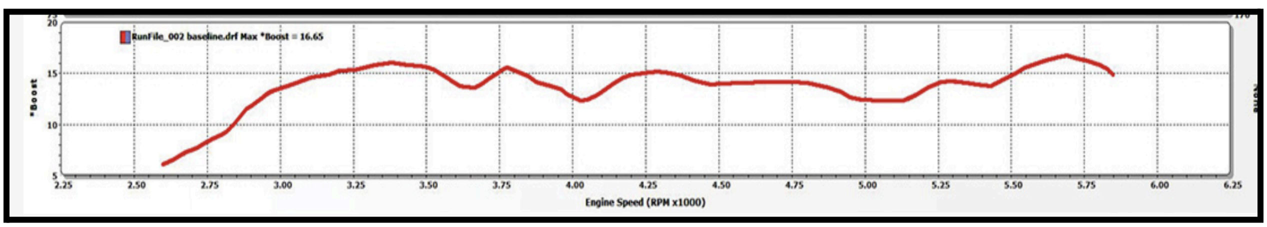
For this study, the following engine speed (N) conditions were measured at 2750, 3000, 3500, 4000, 4500, 5000, and 5500 RPM at each AFR. These values were chosen in order to include a wide range of engine operating conditions. To ensure AFR trend consistency across different speeds, a wide range of RPM values was used alongside an AFR range of 0.8 to 1.2. This specific range was tested in increments of 0.05 to minimize variation, allowing for the exploration of extreme AFR impacts on GTEs not extensively covered in existing research.

AFR

The initial derivation begins with the use of a modified version of the speed density equation, which gives the mass of air flow (m_{air}) necessary for AFR calculations, shown by the following equation:

$$m_{air} = \frac{[\eta_{vol}(N, p_{man})](V_d)(N)(p_{man})}{120(R)(T_{man})} \quad [2, 25]$$

The constants relevant to this equation are volumetric displacement (V_d), ambient pressure (p_{amb}) , and the ideal gas constant (R). The parameters for this equation are the boost pressure (p_{boost}) and engine speed (N). (See Table 2 in Nomenclature) DSPORT Performance, an automotive and car performance magazine, [26] supplied data for the p_{boost} of a 2017 Subaru WRX STI with an EJ257 engine at various engine speeds. The graph is shown below:



Boost Pressure vs RPM [26]
Figure 1.
(Input Data for Equation A1)

The values for Appendix C Table 1 were provided by this figure. The following relationship between p_{boost}, manifold pressure (p_{man}), and p_{amb} can be established by the pressure ratio (p_r) formula Eq. B1:

$$p_r = \frac{p_{man}}{p_{amb}} = \frac{p_{boost}+p_{amb}}{p_{amb}} \qquad [25, 27, 31]$$

where p_{man} can be isolated by using this equation. By solving for p_{man}, we can plug this value back into the modified speed density equation. The V_d of the EJ257 engine was 0.0025 meters cubed from JDM Engine Depot, a reputable car engine auction website [28]. (See Table 2 in Nomenclature) The volumetric efficiency is calculated through Eq. C1 (See Appendix B). This was an empirical equation with inputs of N and p_{man}. However, it's important to note that the volumetric efficiency equation uses the following conversion for corrected engine speed (n):

$$n = \frac{N}{1000} \quad [29]$$

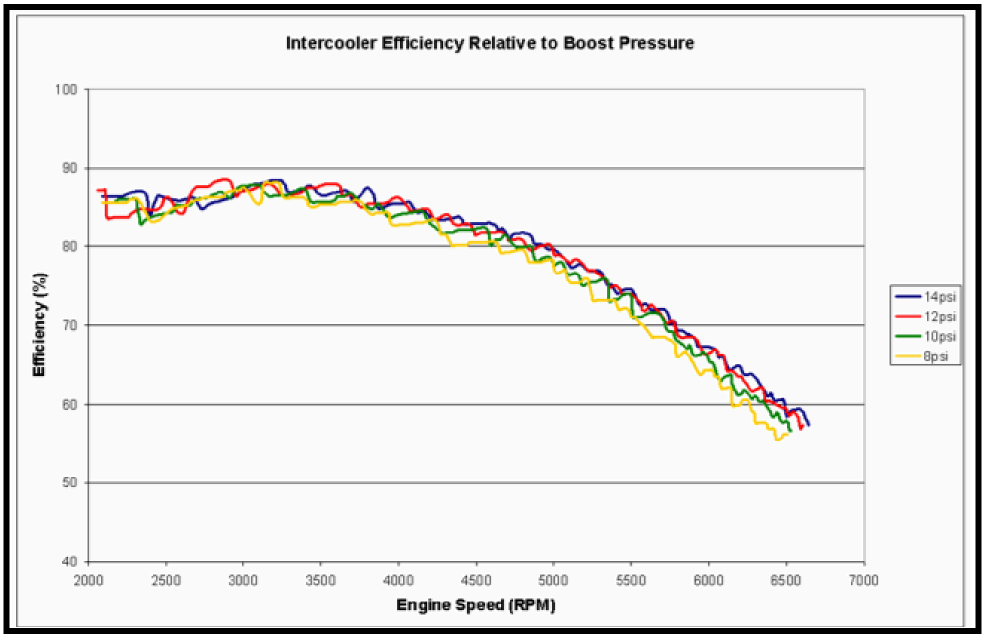
Finally, the temperature in the engine manifold is calculated with Eq. D1-D3:

$$T_{man} = T_c - \epsilon(T_c - T_{cool}) \quad [25]$$

$$\text{(SHR Eq.)} \quad T_c = T_{amb} \left(\frac{p_{man}}{p_{amb}}\right)^{\frac{\gamma-1}{\gamma}} \quad [25]$$

$$T_{cool} = T_{amb} \quad [25]$$

The constants are the ambient temperature (T_{amb}), which equals the temperature after the intercooler (T_{cool}), p_{amb}, and the specific heat ratio (γ), which was initially set equal to 1.4 for isentropic conditions. (See Table 2 in Nomenclature). Note that (SHR Eq.) means it is a specific heat ratio equation, as this value will be replaced by a corrected value afterwards to accurately model knock trends. The parameter for this equation is the intercooler efficiency (ϵ), which was extracted from a website called "Engine Basics," which is a website focused on tuning and repairing engines [30]. The data points were taken with respect to p_{boost} and N.



Intercooler Efficiency vs. RPM [30]
Figure 1.
(Input for Equation D1)

The values for Appendix D Table 1 were provided by this figure.

Now, with the m_{air} calculated, the value can be substituted into Eq. A2 to get AFR:

$$AFR = \frac{m_{air}}{m_{fuel}} \qquad [2, 25]$$

For a more standardized approach for determining rich and lean mixtures, the equation is converted to normalized AFR (λ) in Eq. A3:

$$\lambda = \frac{m_{air}}{14.7\,(m_{fuel})} = \frac{AFR_{actual}}{AFR_{stoichiometric}} \qquad [2]$$

Given the AFR is held constant, m_{fuel} can be calculated.

Torque/BSFC

In order to calculate the power output and fuel efficiency of the GTE, the following formulas were used for calculating torque and brake-specific-fuel consumption in Eq. G1 and Eq. H1 respectively:

$$\tau = \frac{9549\,(BP)}{(N)} \qquad [25, 29]$$

$$BSFC = \frac{(m_{fuel})}{BP} \qquad [8, 32]$$

Both of these formulas were dependent on brake power (BP), which was given by the following in Eq. E1:

$$BP = (\eta_{th})\,(m_{fuel})(Q_{heat}) \qquad [8, 29]$$

(Q_{heat}) is a constant. The total brake thermal efficiency (η_{th}) is given by Eq. F1:

$$\eta_{th} = (\eta_{thn})(\eta_{thp_{man}})(\eta_{th\lambda})(\eta_{th\theta}) \qquad [29, 31]$$

Where n, p_{man}, λ, and spark angle (θ) contribute to thermal efficiency. Using Eq. F1-F9, the values for η_{th} can be calculated. Given that BP is determined by $the\ m_{fuel}$ component and the fact that BP is a component of τ and $BSFC$, AFR plays a role in τ and $BSFC$. This, in combination with the λ contribution to η_{th} further supports this claim.

Knock

To model knock, the residence time (T_{RES}) and ignition delay (T_{ID}) are given by the following Eq. I1 and I2 respectively:

$$T_{RES} = \frac{(13.5 * 1000)}{6N} \qquad [33]$$

$$T_{ID} = 17.69\,\left(\frac{ON}{100}\right)^{3.402} * p_{peak}^{-1.7} * e^{3800/T_{peak}} \qquad [34]$$

Where $T_{ID} < T_{RES}$ means engine knock occurs [33]. Since the peak temperature (T_{peak}) and peak pressure (p_{peak}) were not explicitly stated, Eq. I3 and I4 were substituted into Eq. I1 and I2:

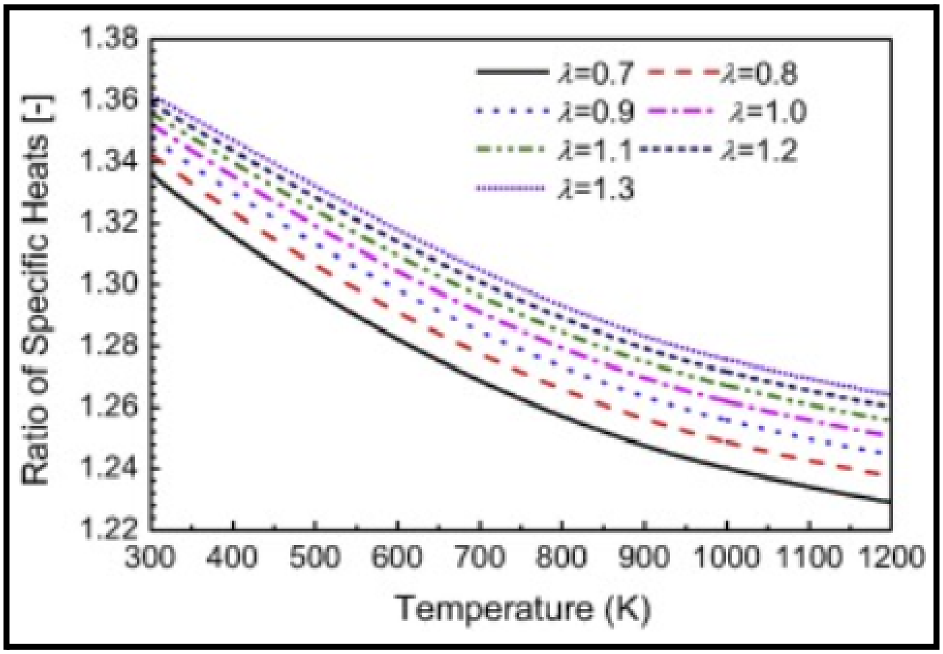
$$\text{(SHR Eq.) } p_{peak} = p_{man} * (CR)^{\gamma} \qquad [35]$$

$$\text{(SHR Eq.) } T_{peak} = T_{man} * \left(\frac{p_{peak}}{p_{man}}\right)^{[1-\frac{1}{\gamma}]} \qquad [35]$$

For an isentropic temperature model, the peak temperature (T_{peak}) is achieved at the end of the compression process, where the piston is at top dead center [35]. This is where the mixture is compressed to its minimum volume [36]. End gas temperature is the temperature in the engine cylinder before the mixture is completely consumed by the propagating flame. Essentially, it's the temperature of the unburned fuel mixture, and it's vital for determining the knock probability in engines. [37] Given that the unburned fuel mixture is inside the combustion chamber at the time its temperature is being recorded combined with the fact that the manifold temperature (T_{man}) is the temperature of the mixture entering the combustion chamber and that (T_{peak}) is the highest operating temperature in the combustion chamber, for the simplified isentropic model, it's reasonable to estimate that the end gas temperature falls between the manifold and peak temperature. A first-order linear approximation of the end gas temperature (T_{EGT}) was obtained by averaging the boundary temperature values:

$$T_{EGT} = \frac{T_{peak} + T_{man}}{2}$$

While these simplifications reduce the quantitative accuracy of the model, they preserve the knock trends for the engine model. These newly calculated T_{EGT} values were used to determine the specific heat ratio at each respective AFR value. The graph that was used is shown below:



Specific Heat Ratio vs. End Gas Temperature at Various AFR [34]
Figure 1.
(Input for Equations D3, I3, and I4)

The values for Appendix E Table 1 were provided by this figure. With these new specific heat ratios, a resubstitution is required for the equations containing specific heat capacity (denoted by SHR equations). The resubstitutions were done for the following equations:

$$T_c = T_{amb} \left(\frac{p_{man}}{p_{amb}}\right)^{\frac{\gamma-1}{\gamma}} \quad [25]$$

$$T_{peak} = T_{man} * \left(\frac{p_{peak}}{p_{man}}\right)^{[1-\frac{1}{\gamma}]} \quad [35]$$

$$p_{peak} = p_{man} * (CR)^{\gamma} \quad [35]$$

With these corrected pressures and temperatures, the ignition delay can be calculated using the original equation:

$$T_{ID} = 17.69\,(\frac{ON}{100})^{3.402} * p_{peak}^{\;-1.7} * e^{3800/T_{peak}} \qquad [34]$$

The knock margin can now be calculated with these corrected values with the following equation:

$$KM = T_{ID} - T_{RES} \text{ (based on [33])}$$

This margin represented the safety score. The higher the knock margin, the safer the engine is from engine knock (less prone to engine damage). The lower the score, the higher the risk of engine knock (more prone to engine damage). Evidently, AFR plays a role in engine knock since it is largely responsible for differing specific heat ratios, which ultimately influence T_{ID} and T_{RES}, components that are directly responsible for engine knock.

Data Processing

This study utilized a min-max normalization in order to give equal weights to all variables to fairly account for each variable in the determination of the optimal AFR range. Using Eq. J1 and J2:

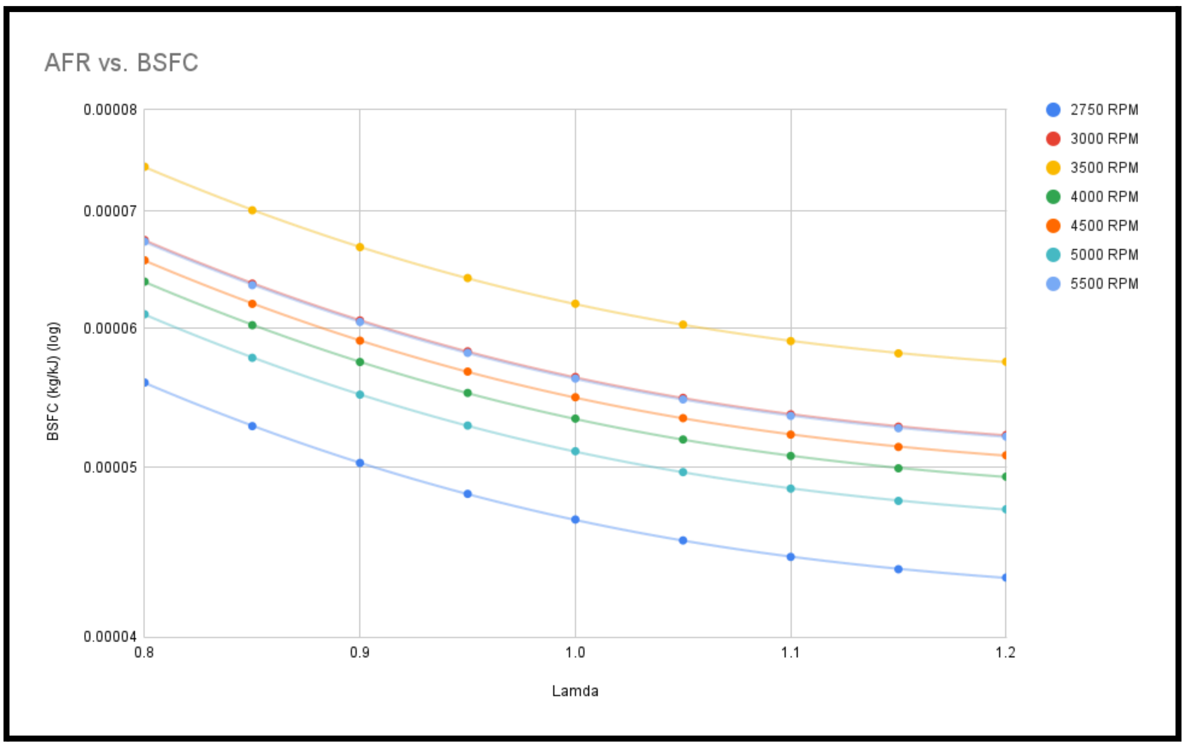
$$Normalized\ Score\ (Z) = \frac{value - min}{max - min} \qquad [38]$$

$$Z_{perf} = Z_{\tau} + Z_{knock} - Z_{BSFC} \qquad [38]$$

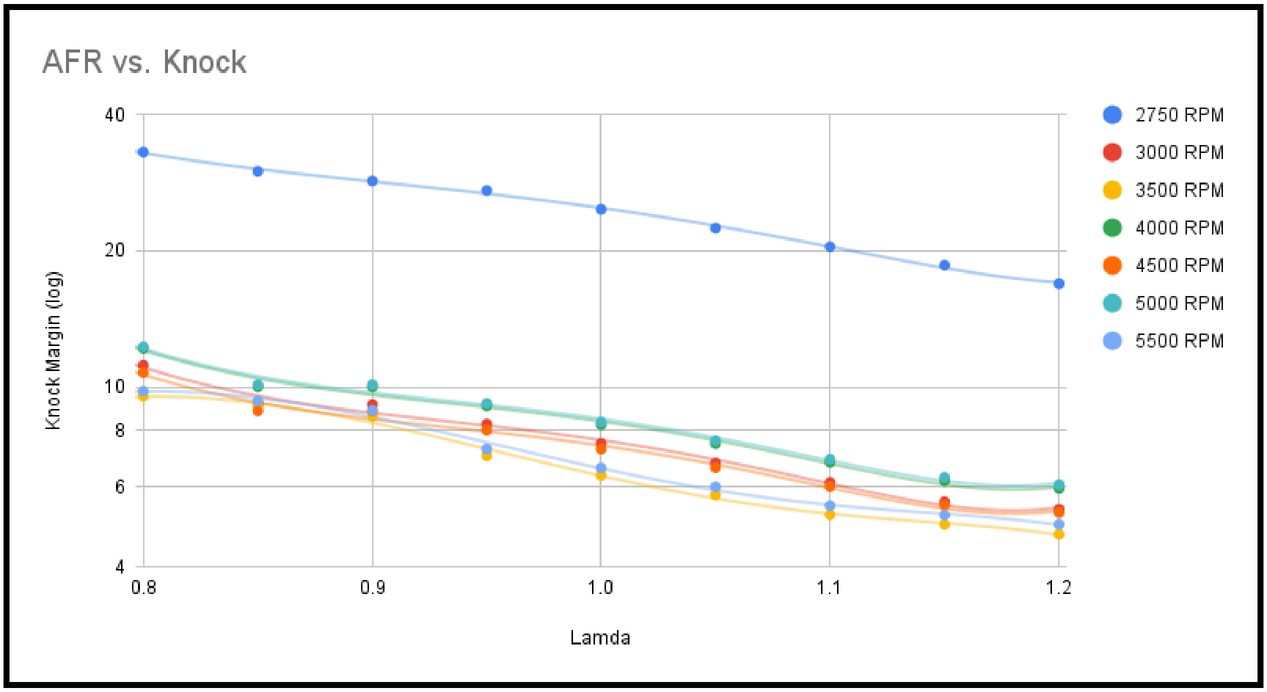
The normalized performance score (Z_{perf}) was the sum of the normalized torque (Z_{τ}) and knock (Z_{knock}) scores (positive contributions) and the BSFC score (Z_{BSFC}) (negative contribution). Z_{τ} and Z_{knock} were added to Z_{perf} because increased power and lower engine damage were ideal for the engine performance. On the contrary, Z_{BSFC} was subtracted because a higher BSFC was worse for the overall fuel economy of the engine. Totaling the AFR value with the highest normalized score presented the most optimal AFR for maximizing torque output while minimizing engine knock and fuel consumption.

Findings and Discussion

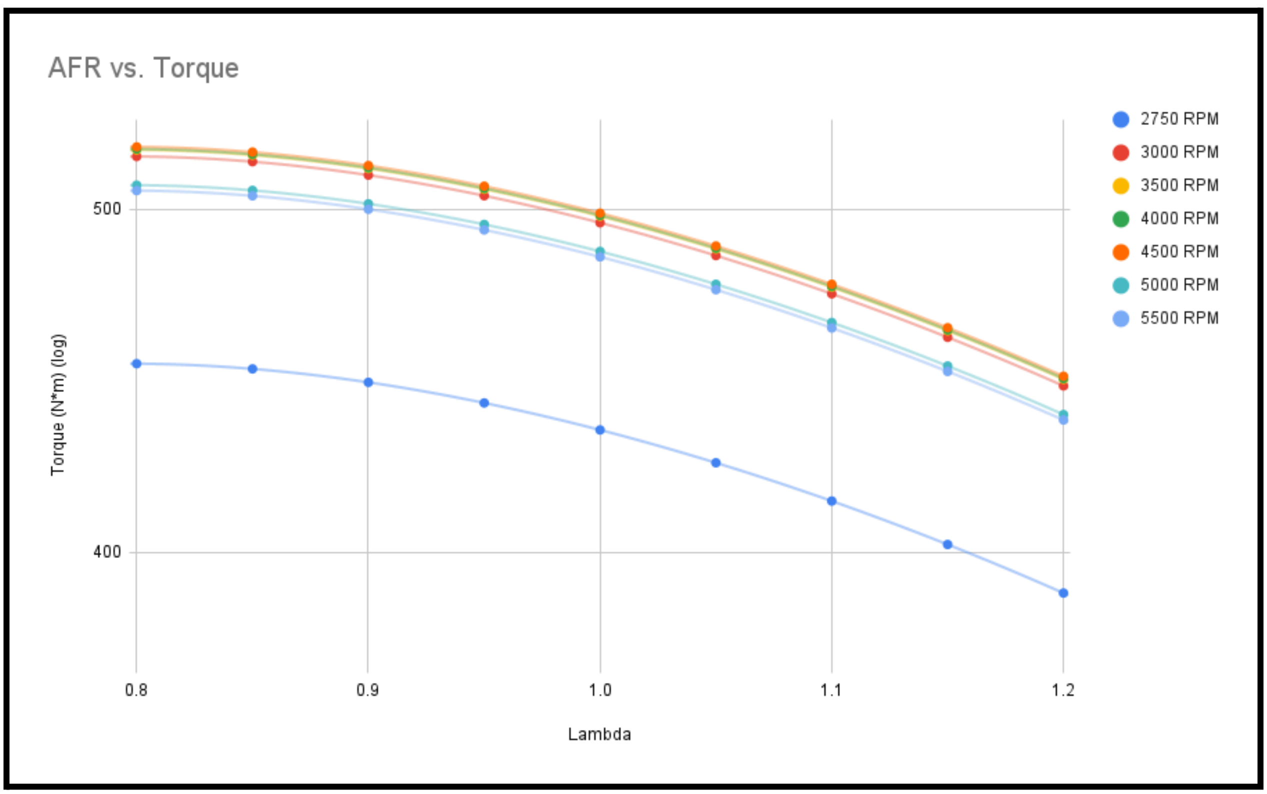
After running the engine model for each respective Normalized Air-to-Fuel Ratio (λ) and Revolutions Per Minute (RPM), all the data was graphed on Google Sheets as seen in figures F1-F3: The relationships between λ, Brake-Specific Fuel Consumption (BSFC), knock, and torque were graphed respectively:



BSFC vs λ Across Different RPM Values
Figure 1: Derived from Appendix F Table 1



Knock Margin vs λ Across Different RPM Values
Figure 2: Derived from Appendix F Table 2



Torque vs λ Across Different RPM Values
Figure 3: Derived from Appendix F Table 3

For all three graphs, the logarithmic trend lines were used instead of linear trend lines in order to accurately reflect the rate of change in the data. After graphing these points, the data was converted to a normalized score, and the following conclusions were drawn: the best λ for torque and knock is 0.8, the best λ for fuel efficiency is 1.2. The λ for the lean best torque output was 0.9. (See Appendix F and G). Results indicate that leaner conditions are better for engine efficiency and that richer conditions are better for increasing torque and reducing engine knock for a Gas Turbocharged Engine (GTE). Additionally, a slightly rich mixture is best for balancing torque output, fuel consumption, and engine knock. The original hypothesis from the literature sources predicted that richer conditions (lower λ) decrease fuel efficiency and knock tendencies, and increase torque output. Additionally, it was predicted that leaner conditions (higher λ) would lead to the inverse of these results. This data supports the general trends; however, the specific λ intervals deviated from initial predictions. This study's data shows that the best λ for maximum torque output and maximum fuel efficiency were the local extrema of the model. This differed from the initial prediction that stated the values would be closer to the stoichiometric ratio. Nonetheless, the data in this study shows that the trends AFR for knock, BSFC, and torque in gas engines generally align with those found in Gas Turbocharged Engines (GTEs).

Conclusion

Implications

The study confirms the general trends that leaner Air-to-Fuel Ratios (AFRs) result in high fuel efficiency at the cost of lower torque output and increased engine damage risk, that lower AFR values result in higher torque output and less engine knock/fuel efficiency, and that the optimal range for operation under various Revolutions Per Minute (RPMs) was a slightly rich mixture. This maximized the power output and fuel efficiency of the Gas Turbocharged Engine (GTE) without risking high engine knock. The optimization strategy in this study, combined with data on engine emissions, could provide engine companies with a simple yet effective way to measure their vehicle efficiency for all sorts of engines, not just GTEs. Whether an engine runs on gas, diesel, or hybrid systems, the optimization methods used in this paper can be easily applied to any engine type with only minor tweaks. The applicability of the GTE optimization in this study encompasses a wide variety of vehicles: submarines, dump truck engines, aircraft, hybrid vehicles, and industrial machinery. Additionally, this simplified model could provide an affordable way of calculating engine performance without the use of external resources. If these companies implement these optimization methods with some methodological refinement, it would help reduce global warming, giving the world more time to ease into the transition from gas and diesel systems to hydrogen and electric systems for industrial machines, planes, and cars alike. This could result in significant improvements in fuel efficiency across numerous applications, which also gives us more time to develop long-term sustainable technologies for the future. To conclude, the results of this study has the potential to reduce fuel consumption, cut spendings on gas in favor of cheaper and more efficient energy sources, and drastically reduce contributions to global warming.

Limitations

Despite the findings aligning with the general trends of the literature review, the data collected from this study were obtained under ideal conditions. Given the lack of access to professional engineering programming software, an engine, and a dynamometer, this study lacked the material resources to generate real engine data. Furthermore, given the limited time frame and computational constraints, creating a hyper-realistic engine model was not feasible.

Many of the studies used for this paper had a team of college researchers working on their engine model for months. While this study was conducted over a similar period, the time commitment and work of one high school researcher could not match the work of a team of college professionals. This study relied heavily on empirical formulas for the total brake thermal efficiency, volumetric efficiency, and ignition delay. In a real-world application, each engine setup has its own unique empirical constants. This was overlooked due to a significant gap in scholarly data for engine performance data on GTEs. Furthermore, the intercooler efficiency data and boost pressure values were derived from secondary sources that were from publicly available enthusiast dynamometer data due to a gap in scholarly data for GTEs at specific RPMs. The same could be said for the T_{EGT} , where there were no scholarly sources establishing the relationship between RPM and end gas temperature. As a result, this study used a first-order linear approximation to generate a qualitative trend that modeled engine knock properties. Finally, the temperature values for peak temperature and end gas temperature were largely dependent on isentropic ideal conditions. Finally, this study was only able to run one trial for one type of engine. The significant lack of dynamometer data from scholarly sources, in combination with the fixed empirical constants of the equation, resulted in a single output for every input. The lack of different engine dynamometer data on GTEs restricted the data to one engine, and the empirical constants allowed for no variance in the data. Given these significant limitations, a real EJ257's engine performance under non-ideal conditions could strongly differ from the results obtained in this study.

Future Research

To further the understanding of the effects of AFR on engine performance, future research could study the effects of AFR on the following gas emissions: HC, NOx, and CO to directly address AFR’s effects on global warming. Additionally, future research could focus more on the relationship between RPM, end gas temperature, intercooler efficiency, and boost pressure and their effects on AFR and GTE performance. The relationship between these variables and AFR was not directly explored as extensively as other variables by other scholarly sources. The engine optimization strategies in this study can be applied to other engine types, such as diesel, hybrid, hydrogen, and naturally aspirated gas engines, in order to improve the efficiency of other vehicles. Furthermore, this study’s data were limited by the RPM ranges of publicly available enthusiast dynamometer data, so future research should include lower RPM ranges in order to explore the effects of AFR on GTE performance for lower speed conditions. Finally, the experiment should be run under multiple engine setups (V4s, V6s, and V8s) in order to see if the general trends align for different engine type configurations to increase the result's generalizability.

References

[1] A. Isenstadt, J. German, M. Dorobantu, D. Boggs, and T. Watson, “Downsized, boosted gasoline engines,” 2016. Available: https://theicct.org/sites/default/files/publications/Downsized-boosted-gasoline-engines_working-paper_ICCT_27102016.pdf

[2] A. Nemati, I. Jebellat, and S. Nasiri, “Experimental Air-Fuel Ratio Control to Improve Emission Performance using Smooth Super-Twisting Sliding Mode Control in a Gasoline Engine,” *SSRN*, 2024, doi: https://doi.org/10.2139/ssrn.4975873.

[3] A. Foley, “Renewable and Sustainable Energy Reviews | ScienceDirect.com,” *Sciencedirect.com*, 2019. https://www.sciencedirect.com/journal/renewable-and-sustainable-energy-reviews

[4] Jing Na *et al.*, “Air–Fuel Ratio Control of Spark Ignition Engines With Unknown System Dynamics Estimator: Theory and Experiments,” *IEEE*, vol. 29, no. 2, pp. 786–793, Mar. 2021, doi: https://ieeexplore.ieee.org/document/8928941

[5] E. Frasci, R. Novella Rosa, B. Plá Moreno, I. Arsie, and E. Jannelli, “Impact of prechamber design and air–fuel ratio on combustion and fuel consumption in a SI engine equipped with a passive TJI,” *Fuel*, vol. 345, p. 128265, Aug. 2023, doi: https://doi.org/10.1016/j.fuel.2023.128265.

[6] S. P. Chincholkar and J. G. Suryawanshi, "Gasoline Direct Injection: An Efficient Technology," *Energy Procedia*, vol. 90, pp. 666–672, Dec. 2016, doi: https://doi.org/10.1016/j.egypro.2016.11.235.

[7] A. Roy, B. Y. McCabe, S. Saxe, and I. Daniel Posen, "Review of factors affecting earthworks greenhouse gas emissions and fuel use," *Renewable & sustainable energy reviews*, vol. 194, pp. 114290–114290, Apr. 2024, doi: https://doi.org/10.1016/j.rser.2024.114290.

[8] E. Shamsher Khan, "Si Engine Effect of Air Fuel Ratio on Engine Performance and Emission With Ethanol Blends (Cdi+efi)," *Research Square*, Jan. 21, 2023. https://www.researchgate.net/publication/367343071_Si_Engine_Effect_of_Air_Fuel_Ratio_on_Engine_Performance_and_Emission_With_Ethanol_Blends_Cdiefi?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6InB1YmxpY2F0aW9uIiwicGFnZSI6Il9kaXJlY3QifX0

[9] M. Amann, D. Mehta, and T. Alger, "Engine Operating Condition and Gasoline Fuel Composition Effects on Low-Speed Pre-Ignition in High-Performance Spark Ignited Gasoline Engines," *SAE International Journal of Engines*, vol. 4, no. 1, pp. 274–285, 2011, doi: https://www.jstor.org/stable/26278150?read-now=1&seq=1#page_scan_tab_contents

[10] P. Andersson, "Air Charge Estimation in Turbocharged Spark Ignition Engines," 2005. Available: https://www.fs.isy.liu.se/en/Publications/PhD/05_PhD_989_PA.pdf

[11] S. Miles, "Turbochargers – USC Viterbi School of Engineering," *Illumin Magazine*, 2005. https://illumin.usc.edu/turbochargers/

[12] A. Robert, S. Richard, O. Colin, L. Martinez, and Loic de Francqueville, "LES prediction and analysis of knocking combustion in a spark ignition engine," *Proceedings of the Combustion Institute*, vol. 35, no. 3, pp. 2941–2948, Jan. 2015, doi: https://doi.org/10.1016/j.proci.2014.05.154.

[13] T. Wagner, "Advanced Gasoline Turbocharged Direct Injection (GTDI) Engine Development," *OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)*, Dec. 2015, doi: https://doi.org/10.2172/1253890.

[14] Renewable and Sustainable Energy Reviews, "Air-Fuel Ratio - an overview | ScienceDirect Topics," *www.sciencedirect.com*, 2019. https://www.sciencedirect.com/topics/engineering/air-fuel-ratio

[15] J. G. Speight, *Handbook of Industrial Hydrocarbon Processes*. Amsterdam ; Boston: Elsevier/Gulf Professional Publishing, 2011. https://www.sciencedirect.com/book/monograph/9780750686327/handbook-of-industrial-hydrocarbon-processes

[16] Q. Xin, "Diesel Engine System Design," *ScienceDirect*, 2025.

https://www.sciencedirect.com/book/monograph/9781845697150/diesel-engine-system-design (accessed Nov. 24, 2025).

[17] P. Geng, E. Cao, Q. Tan, and L. Wei, "Effects of alternative fuels on the combustion characteristics and emission products from diesel engines: A review," *Renewable and Sustainable Energy Reviews*, vol. 71, pp. 523–534, May 2017, doi: https://doi.org/10.1016/j.rser.2016.12.080.

[18] R. N. Brady, *Reference Module in Earth Systems and Environmental Sciences*. Elsevier, 2013.https://www.sciencedirect.com/referencework/9780124095489/earth-systems-and-environmental-sciences

[19] T. and F. Staff, "WOT - Knowledge and References | Taylor & Francis," *Taylor & Francis*, 2021.https://taylorandfrancis.com/knowledge/Engineering_and_technology/Mechanical_engineering/WOT/

[20] S. T. P. Purayil, M. O. Hamdan, S. A. B. Al-Omari, M. Y. E. Selim, and E. Elnajjar, "Review of hydrogen–gasoline SI dual fuel engines: Engine performance and emission," *Energy Reports*, vol. 9, pp. 4547–4573, Dec. 2023, doi: https://doi.org/10.1016/j.egyr.2023.03.054.

[21] A. Almaleki, P. Hellier, N. Ladommatos, M. Talibi, and Z. Khan, "Effects of fuel composition at varying air-fuel ratio on knock resistance during spark-ignition combustion," *Fuel*, vol. 344, p. 128015, Jul. 2023, doi: https://doi.org/10.1016/j.fuel.2023.128015.

[22] C. W. Wu, R. H. Chen, J. Y. Pu, and T. H. Lin, "The influence of air-fuel ratio on engine performance and pollutant emission of an SI engine using ethanol-gasoline-blended fuels," *Atmospheric Environment*, vol. 38, no. 40 SPEC.ISS., pp. 7093–7100, Dec. 2004, doi: https://doi.org/10.1016/j.atmosenv.2004.01.058.

[23] US EPA,OAR,OAP,CCD, "Install Automated Air/Fuel Ratio Controls | EPA's Voluntary Methane Programs for the Oil and Natural Gas Industry | US EPA," *Epa.gov*, 2016. https://19january2017snapshot.epa.gov/sites/production/files/2016-06/documents/auto-air-fuel-ratio.pdf (accessed Dec. 04, 2025).

[24] Mohammadjavad Nekooei, Jaswar Koto, A. Priyanto, and Z. Dehghani, "A Review on Engine MVEM Models and AFR Control Methods to Developing a New Engine Model and Ziegler...," *ResearchGate*, Aug. 25, 2013. https://www.researchgate.net/publication/317798049_A_Review_on_Engine_MVEM_Models_and_AFR_Control_Methods_to_Developing_a_New_Engine_Model_and_Ziegler_Nichols_PID_Fuzzy_controller_Applied_to

[25] K. Rong, "MODELING OF TURBOCHARGED SPARK-IGNITED ENGINE AND

MODEL PREDICTIVE CONTROL OF HYBRID TURBOCHARGER," 2014. Accessed: Dec. 15, 2025. [Online]. Available: https://ufdcimages.uflib.ufl.edu/uf/e0/04/67/95/00001/rong_k.pdf

[26] M. Ferrara, "Test & Tune: 2017 Subaru WRX STI | Part:1 Intake and Tune Beefs Up the EJ257 - DSPORT Magazine," *DSPORT Magazine*, Jun. 29, 2017. https://dsportmag.com/the-tech/test-n-tune/test-tune-2017-subaru-wrx-sti-intake-and-tune-beefs-up-the-ej257/ (accessed Jan. 12, 2026).

[27] G. Motion Company, "Turbo Tech 103 (Expert)," 2017. Available: https://www.garrettmotion.com/wp-content/uploads/2018/06/Turbo-Tech-103.pdf

[28] "EJ257 Engine for Sale," *JDM Engine Depot*, 2018. https://www.jdmenginedepotinc.com/ej257-engine-for-sale/?srsltid=AfmBOopxU311yxE7-KrpNfdXRAFqLrDsh5I3_GtdlwaDOZ81l28Rm-X1 (accessed Feb. 03, 2026).

[29] M. Saeedi, "A Mean Value Internal Combustion Engine Model in MapleSim," *Uwaterloo.ca*, Aug. 31, 2010. https://uwspace.uwaterloo.ca/items/e0a88ac8-4815-481b-8dd1-deed2c687c1d

[30] Engine Basics, "Front Mount Intercooler Efficency Test Data," *Enginebasics.com*, 2025. https://www.enginebasics.com/Advanced%20Engine%20Tuning/Intercooler%20Efficency%20Test%20.html

[31] E. Hendricks and S. C. Sorenson, “Mean Value SI Engine Model for Control Studies,” *American Control Conference*, pp. 1882–1887, 1990, Accessed: Jan. 26, 2026. [Online]. Available: https://orbit.dtu.dk/en/publications/mean-value-si-engine-model-for-control-studies/

[32] S. M. A. Rahman, H. H. Masjuki, M. A. Kalam, A. Sanjid, and M. J. Abedin, “Assessment of emission and performance of compression ignition engine with varying injection timing,” *Renewable and Sustainable Energy Reviews*, vol. 35, pp. 221–230, Jul. 2014, doi: https://doi.org/10.1016/j.rser.2014.03.049.

[33] A. Lewis, E. Ortiz-Soto, G. Lavoie, and D. N. Assanis, “Scaling and dimensional methods to incorporate knock and flammability limits in models of high-efficiency gasoline and ethanol engines,” *International Journal of Engine Research*, vol. 16, no. 2, Jan. 2014, doi: https://www.researchgate.net/publication/268223144_Scaling_and_dimensional_methods_to_incorporate_knock_and_flammability_limits_in_models_of_high-efficiency_gasoline_and_ethanol_engines

[34] L.-L. Chen, T. Li, T. Yin, and B. Zheng, “A predictive model for knock onset in spark-ignition engines with cooled EGR,” *Sci Direct*, vol. 87, pp. 946–955, Nov. 2014, doi: https://www.sciencedirect.com/science/article/abs/pii/S0196890414007262.

[35] NASA, “Isentropic Compression or Expansion,” *Nasa.gov*, 2015. https://www.grc.nasa.gov/WWW/K-12/airplane/compexp.html

[36] Florida State University, “Gas Laws,” *Florida State University*, 2019. https://www.chem.fsu.edu/chemlab/chm1045/gas_laws.html

{37] S. Gersen, M. van Essen, H. Levinsky, and G. van Dijk, “SAE International | Advancing mobility knowledge and solutions,” *Sae.org*, 2026. https://www.sae.org/articles/characterizing-gaseous-fuels-knock-resistance-based-chemical-physical-properties-fuel-2015-01-9077 (accessed Feb. 17, 2026).

[38] G. Ciaburro, V. K. Ayyadevara, A. Perrier, B. Fry, and A. Gulli, “Min–max normalization - Hands-On Machine Learning on Google Cloud Platform [Book],” *www.oreilly.com*, Apr. 2018. https://www.oreilly.com/library/view/hands-on-machine-learning/9781788393485/fd5b8a44-e9d3-4c19-bebb-c2fa5a5ebfee.xhtml

[39] Google, “Gemini,” *gemini.google.com*, 2025. https://gemini.google.com/app

Appendices

AI Disclosure and Usage Statement

During the preparation of this paper, Gemini 3 Pro was used to help interpret complex equations and understand the underlying principles of the created engine model. Given the significant lack of resources, whether it be mentorship, articles, or an engine, I validated my equations through AI use in order to ensure the accuracy of my equations. I cross-checked its output with the scholarly articles I found during my research in order to validate its claims and accuracy. Such uses included interpreting the meaning of the spark variables for the generation of the brake thermal efficiency and the interpretation of the manifold temperature variables. Additionally, I double-checked my method of generating end gas temperatures for qualitative engine knock trends. Having found a source that showed the relationship between end gas temperature, specific heat ratios, and AFR, I needed to convert my values of peak temperature and manifold temperature into an end gas temperature. I was more concerned about graphing the general trend for engine knock rather than focusing on the quantitative accuracy of my values, so I figured that I could take the average between the peak temperature and manifold temperature. Logically, this made sense; the engine temperature would no doubt be sandwiched between the peak temperature of the chamber and the temperature at which the air-fuel mixture entered the chamber. I double checked this with Gemini in order to ensure this was a mathematically valid approach for estimating the qualitative trend for knock, which complemented other studies backing this trend. Furthermore, when looking for definitions in my literature review, I found many SciDirect webpages with generalized definitions for certain terms. They had an AI-generated definition because there was no explicitly stated definition, and the authors assumed that the reader already had a strong background on the topic. The other sources that did explicitly mention the definition were non-academic sources, so I used the AI-generated

SciDirect definitions in this paper. I can fully verify that I found and read all sources on my own, performed the necessary analysis for this paper, verified the credibility and accuracy of my sources, and made decisions on how to communicate effectively in this paper.

Appendix A

The following conversions are used for calculating the values of volumetric efficiency, total brake thermal efficiency, and knock:

This equation was used for converting the pressure values.

$$100,000\ Pascals = 1\ bar$$

This equation was used to scale the RPM.

$$n = \frac{N}{1000}$$ [29]

1. Deriving equation for $m_{air} = \frac{(\eta_{vol})(V_d)(N)(p_{man})}{120(R)(T_{man})}$
   a. Constants
      i. p_{man} is calculated from p_{boost} (parameter) [26] (See Appendix C Table 1)
         1. $p_r = \frac{p_{man}}{p_{amb}} = \frac{p_{boost}+p_{amb}}{p_{amb}}$ [25, 27, 31]
         2. Convert PSI to pascals.
         3. 1 PSI = 6894.7 Pa
      ii. R = 287 [25]
      iii. N = 2750 (initial RPM condition)
      iv. V_d = 0.0025 meters cubed [28]
   b. Variables
      i. η_{vol}

1. $100,000Pa = 1\ bar$ (equation must be in bar)

2. $\frac{N}{1000} = n$ (RPM conversion for this formula) [29]

3. $\eta_{vol} = 0.696 + 0.16p_{man} - 0.0262n - 0.00643n^2$ [29]

ii. T_{man}

1. $T_{man} = T_c - \epsilon(T_c - T_{cool})$ [25]

2. ϵ (parameter) [30] (See Appendix D Table 1)

3. $T_{cool} = T_{amb} = 296$ [25]

4. $T_c = T_{amb} (\frac{p_{man}}{p_{amb}})^{\frac{\gamma-1}{\gamma}}$ $\gamma = 1.4$ (initially) [25]

c. $AFR = \frac{m_{air}}{m_{fuel}}$ (can set AFR to a value like 14.7 and we can get $m_{fuel} = \frac{m_{air}}{14.7}$ [2, 25]

2. Torque and BSFC depend on the Brake Power so solve for Brake Power

a. $BP = (\eta_{th})(m_{fuel})(Q_{heat})$ [8, 29]

i. Constants

1. $Q_{heat} = 44,000$ [25]

2. m_{fuel} is already calculated

ii. Variables

1. Calculate η_{th}

a. $\eta_{th} = (\eta_{thn})(\eta_{thp_{man}})(\eta_{th\lambda})(\eta_{th\theta})$ [29, 31]

b. $100,000Pa = 1\ bar$ (equation must be in bar)

c. $\frac{N}{1000} = n$ (RPM conversion for this formula) [29]

i. $\eta_{thn} = 0.558[1 + (-0.392)n^{-0.36}]$ [29]

ii. $\eta_{thp_{man}} = 0.9301 + 0.2154p_{man} - 0.1657(p_{man})^2$ [29]

iii. $\eta_{th\lambda} = -1.299 + 3.599\lambda - 1.332\lambda^2 - 0.0205 + 1.741\lambda - 0.745\lambda^2$ [29]

iv. $\eta_{th\theta} = 0.7 + 0.024(\theta - \theta_{mbt}) - 0.00048(\theta - \theta_{mbt})^2$ [29]

v. $\theta_1 = 47.31p_{man} + 2.6214 + n_{47}$ [29]

vi. $\theta_2 = -56.55p_{man} + 57.34 + n_{47}$ [29]

vii. $\theta_{mbt} = min(\theta_1, \theta_2)$ [29]

1. $n_{47} = 4.7$n $if\ n < 4.8$ & $= 21.62$ $if\ n > 4.8$ [29]
2. $\theta_{mbt} = \theta$ assumes perfect spark advance with no retardation or advancing

b. Input into the following equations for Torque and BSFC

i. $\tau = \frac{9549\ (BP)}{(N)}$ [25, 29]

ii. $BSFC = \frac{(m_{fuel})}{BP}$ [8, 32]

3. Calculate Knock

a. $100,000Pa = 1\ bar$ (equation must be in bar)

b. Constants

i. ON = 92 [33]

ii. $\gamma = 1.4$ (initially) [25]

iii. CR = 8.71 [28]

c. Variables

i. $T_{ID} < T_{RES}$ means knock [33]

ii. $T_{RES} = \frac{(13.5 * 1000)}{6N}$ [33]

iii. $T_{ID} = 17.69\,(\frac{ON}{100})^{3.402} * p_{peak}^{-1.7} * e^{3800/T_{peak}}$ [34]

1. $T_{peak} = T_{man} * (\frac{p_{peak}}{p_{man}})^{[1-\frac{1}{\gamma}]}$ [35]

2. $p_{peak} = p_{man} * (CR)^{\gamma}$ [35]

iv. Using the formulas above, substitute each respective variable into each respective equation above it.

4. Get the average temperature of the engine gas temperatures

a. $T_{EGT} = \frac{T_{peak}+T_{man}}{2}$

5. Use this temperature to find γ at specific AFR values using a graph to find parameters [34] (See Appendix E Table 1)
6. Plug the new γ back into the each equation that has γ in it:
    a. $T_c = T_{amb} \left(\frac{p_{man}}{p_{amb}}\right)^{\frac{\gamma-1}{\gamma}}$ [25]
    b. $p_{peak} = p_{man} * (CR)^{\gamma}$ (calculated in units of bar to remove step of unit conversion) [35]
    c. $T_{peak} = T_{man} * \left(\frac{p_{peak}}{p_{man}}\right)^{[1-\frac{1}{\gamma}]}$ [35]
7. With these updated values, plug them back into
    a. $T_{ID} = 17.69 \left(\frac{ON}{100}\right)^{3.402} * p_{peak}^{-1.7} * e^{3800/T_{peak}}$ [34]
    b. Use the following formula to calculate the safety score
        i. $T_{ID} - T_{RES} = Knock\ Margin$ (based on [33])
8. Repeat steps 1-8 for each respective value of N
    a. 3000
    b. 3500
    c. 4000
    d. 5000
    e. 5500
9. Process the data
    a. Convert the data from knock margin, $BSFC$, and torque into normalized scores using the max-min normalization method.
        i. $Z = \frac{value-min}{max-min}$ (gives score between 0 and 1) [38]

ii. To calculate the best AFR for each respective variable

a. Get the largest score for Torque and Knock

b. Get the smallest score for the absolute value of BSFC

iii. Weighted Sum Model for AFR of Best Performance

1. $Performance\ Score\ Z_P = Z_\tau + Z_{knock} - Z_{BSFC}$ [38]

Appendix B

Air to Fuel Ratio

A1) $$m_{air} = \frac{[\eta_{vol}(N, p_{man})](V_d)(N)(p_{man})}{120(R)(T_{man})}$$ [2, 25]

A2) $$AFR = \frac{m_{air}}{m_{fuel}}$$ [2, 25]

A3) $$\lambda = \frac{m_{air}}{14.7\ (m_{fuel})} = \frac{AFR_{actual}}{AFR_{stoichiometric}}$$ [2]

Pressure

B1) $$p_r = \frac{p_{man}}{p_{amb}} = \frac{p_{boost}+p_{amb}}{p_{amb}}$$ [25, 27, 31]

Volumetric Efficiency (bar and corrected RPM)

C1) $$\eta_{vol} = 0.696 + 0.16p_{man} - 0.0262n - 0.00643n^2$$ [29]

Manifold Temperature (Kelvin)

D1) $$T_{man} = T_c - \epsilon(T_c - T_{cool})$$ [25]

D2) $$T_{cool} = T_{amb}$$ [25]

D3) $$T_c = T_{amb}\left(\frac{p_{man}}{p_{amb}}\right)^{\frac{\gamma-1}{\gamma}}$$ [25]

Brake Power (bar and corrected RPM)

E1) $$BP = (\eta_{th})(m_{fuel})(Q_{heat})$$ [8, 29]

Brake Thermal Efficiency (bar and corrected RPM)

F1) $\eta_{th} = (\eta_{thn})(\eta_{thp_{man}})(\eta_{th\lambda})(\eta_{th\theta})$ [29, 31]

F2) $\eta_{thn} = 0.558[1 + (-0.392)n^{-0.36}]$ [29]

F3) $\eta_{thp_{man}} = 0.9301 + 0.2154p_{man} - 0.1657(p_{man})^2$ [29]

F4) $\eta_{th\lambda} =- 1.299 + 3.599\lambda - 1.332\lambda^2 - 0.0205 + 1.741\lambda - 0.745\lambda^2$ [29]

F5) $\eta_{th\theta} = 0.7 + 0.024(\theta - \theta_{mbt}) - 0.00048(\theta - \theta_{mbt})^2$ [29]

F6) $n_{47} = 4.7n \;\; if\, n < 4.8 \quad \& \quad = 21.62\, if\, n > 4.8$ [29]

F7) $\theta_1 = 47.31p_{man} + 2.6214 + n_{47}$ [29]

F8) $\theta_2 =- 56.55p_{man} + 57.34 + n_{47}$ [29]

F9) $\theta_{mbt} = min(\theta_1, \theta_2)$ [29]

Torque

G1) $\tau = \frac{9549\,(BP)}{(N)}$ [25, 29]

Brake Specific Fuel Consumption

H1) $BSFC = \frac{(m_{fuel})}{BP}$ [8, 32]

Knock (bar)

I1) $T_{RES} = \frac{(13.5 * 1000)}{6N}$ [33]

I2) $T_{ID} = 17.69\,(\frac{ON}{100})^{3.402} * p_{peak}^{-1.7} * e^{3800/T_{peak}}$ [34]

I3) $T_{peak} = T_{man} * (\frac{p_{peak}}{p_{man}})^{[1-\frac{1}{\gamma}]}$ [35]

I4) $p_{peak} = p_{man} * (CR)^{\gamma}$ [35]

Data Processing

J1) $Normalized\ Score\ (Z) = \frac{value-min}{max-min}$ [38]

J2) $Z_{perf} = Z_{\tau} + Z_{knock} - Z_{BSFC}$ [38]

Appendix C

RPM	**2750**	**3000**	**3500**	**4000**	**4500**	**5000**	**5500**
p_{boost} **(psi)**	7	13.5	16	13	14	12.5	15

Table 1: Boost Pressure vs. RPM [26]

Appendix D

RPM	**2750**	**3000**	**3500**	**4000**	**4500**	**5000**	**5500**
Intercooler Efficiency (ϵ)	0.85	0.85	0.87	0.86	0.85	0.79	0.76

Table 1: Intercooler Efficiency vs. RPM [30]

Appendix E

Lambda	**2750 RPM**	**3000 RPM**	**3500 RPM**	**4000 RPM**	**4500 RPM**	**5000 RPM**	**5500 RPM**
0.8	1.28	1.29	1.285	1.29	1.29	1.29	1.285
0.85	1.285	1.3	1.2875	1.3	1.3	1.3	1.2875
0.9	1.2875	1.3	1.29	1.3	1.3	1.3	1.29
0.95	1.29	1.305	1.3	1.305	1.305	1.305	1.3
1	1.295	1.31	1.305	1.31	1.31	1.31	1.305

1.05	1.3	1.315	1.31	1.315	1.315	1.315	1.31
1.1	1.305	1.32	1.315	1.32	1.32	1.32	1.315
1.15	1.31	1.325	1.3175	1.325	1.325	1.325	1.3175
1.2	1.315	1.327	1.32	1.327	1.327	1.327	1.32

Table 1: Specific Heat Ratio vs. RPM and AFR [34]

Appendix F

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM
0.8	0.00005588735705	0.0000674047538	0.00007422622948	0.00006381895584	0.00006563558885	0.00006113916365	0.00006727289349
0.85	0.00005277746872	0.00006365397244	0.00007009586266	0.00006026770853	0.00006198325382	0.00005773703512	0.00006352944957
0.9	0.00005028322555	0.00006064570983	0.00006678315876	0.00005741947947	0.00005905394873	0.00005500840473	0.00006052707187
0.95	0.0000482794644	0.00005822900892	0.00006412188361	0.00005513134221	0.0000567006787	0.00005281634758	0.00005811509862
1	0.00004667737366	0.00005629675558	0.00006199408295	0.00005330188088	0.000054819141	0.00005106370631	0.00005618662525
1.05	0.00004541306323	0.00005477189311	0.00006031490181	0.00005185813805	0.00005333430142	0.00004968058701	0.00005466474579
1.1	0.00004444037746	0.00005359875398	0.00005902303901	0.00005074740757	0.00005219195354	0.00004861649671	0.00005349390161
1.15	0.00004372627214	0.00005273748419	0.00005807460724	0.0000499319556	0.0000513532894	0.000047835286	0.00005263431666
1.2	0.00004324781215	0.00005216042204	0.00005743914542	0.000049385592	0.00005079137333	0.0000473118645	0.00005205838339

Table 1: BSFC vs. λ Values

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM
0.8	33.04640881	11.16173405	9.535821741	12.16536309	10.75777236	12.2756999	9.786922232
0.85	29.96607399	9.137286085	9.063152467	10.00246154	8.84438769	10.11257227	9.313158332
0.9	28.54369349	9.137286085	8.615062593	10.00246154	8.84438769	10.11257227	8.86407732
0.95	27.19400435	8.272443731	7.041670919	9.078326757	8.027081295	9.188491268	7.287572349

1	24.69675742	7.491858845	6.36979942	8.24415147	7.289446867	8.354443598	6.614548862
1.05	22.44478194	6.786551514	5.76352609	7.490361799	6.622984475	7.600834309	6.007328376
1.1	20.41193617	6.148575554	5.215831627	6.808483886	6.020174006	6.919172826	5.458853597
1.15	18.57506922	5.570891189	4.96198293	6.191008712	5.474354285	6.301936951	5.204667325
1.2	16.91365677	5.355275971	4.720509135	5.960533138	5.27063741	6.071561106	4.962886424

Table 2: Knock Margin vs. λ Values

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM
0.8	452.2465573	517.4390044	519.6426999	519.9512061	520.6704984	507.8773196	506.0836797
0.85	450.724687	515.6977528	517.8940325	518.2015006	518.9183724	506.1682443	504.3806402
0.9	446.8000265	511.2073429	513.3844987	513.6892895	514.3999191	501.7608121	499.9887735
0.95	440.8519637	504.4018524	506.5500246	506.8507578	507.5519272	495.0810793	493.3326311
1	433.1840091	495.6285433	497.7393513	498.0348538	498.7238273	486.469891	484.7518544
1.05	424.0418614	485.1685326	487.234793	487.524059	488.1984921	476.2031695	474.5213912
1.1	413.6265469	473.2518251	475.267334	475.5494951	476.2073627	464.5066692	462.8661988
1.15	402.1041309	460.0684247	462.0277874	462.3020883	462.9416297	451.566883	449.9721113
1.2	389.6130011	445.776668	447.6751642	447.9409441	448.5606186	437.5392217	435.9939906

Table 3: Torque vs. λ Values

Appendix G

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM	SUM
0.8	-0.4080113234	-0.7797990952	-1.000000005	-0.6640476051	-0.7226894972	-0.5775424696	-0.7755425731	-4.927632568
0.85	-0.3076224526	-0.6587218521	-0.8666695388	-0.5494114289	-0.6047901549	-0.467719928	-0.6547021869	-4.109637542
0.9	-0.2271069395	-0.5616135144	-0.7597336722	-0.4574690604	-0.5102306087	-0.3796382688	-0.5577838171	-3.453575881

0.95	-0.1624244502	-0.483601107	-0.6738262767	-0.383606757	-0.4342657843	-0.3088774831	-0.4799240213	-2.92652588
1	-0.110708098	-0.4212269284	-0.6051397251	-0.3245507598	-0.3735287337	-0.252301279	-0.4176718619	-2.505127386
1.05	-0.06989547626	-0.3720035465	-0.5509348533	-0.2779459633	-0.3255973123	-0.2076534428	-0.3685447731	-2.172575367
1.1	-0.03849665537	-0.3341339835	-0.5092328261	-0.2420909852	-0.2887217074	-0.1733040345	-0.3307492922	-1.916729484
1.15	-0.01544495119	-0.3063317304	-0.4786169377	-0.2157677563	-0.2616491769	-0.1480861321	-0.3030014272	-1.728898112
1.2	0	-0.2877038533	-0.4581038878	-0.198130845	-0.2435102236	-0.1311898042	-0.2844099909	-1.603048605

Table 1: Normalized BSFC Score Values

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM	SUM
0.8	1	0.2273970107	0.1699968107	0.262828508	0.2131357977	0.2667237704	0.1788615068	2.318943404
0.85	0.8912537692	0.155927155	0.1533099877	0.186470773	0.1455868517	0.1903580539	0.1621360398	1.88504263
0.9	0.8410389301	0.155927155	0.1374909006	0.186470773	0.1455868517	0.1903580539	0.1462819621	1.803154626
0.95	0.7933903413	0.1253952967	0.08194485648	0.1538456914	0.1167331734	0.157734871	0.0906260081	1.519670238
1	0.705229084	0.0978380119	0.05822552164	0.1243964843	0.0906921849	0.1282901692	0.06686600419	1.27153746
1.05	0.625726738	0.07293827918	0.03682202389	0.09778516112	0.06716380983	0.1016852141	0.04542906863	1.047550295
1.1	0.5539604113	0.05041557142	0.01748655813	0.07371256604	0.04588256278	0.07762025976	0.02606605509	0.8451439845
1.15	0.4891127994	0.03002136079	0.008524841139	0.05191360534	0.02661328178	0.05582974709	0.01709242054	0.6791080561
1.2	0.4304593242	0.02240941479	0	0.04377703856	0.01942138752	0.04769670112	0.008556737532	0.5723206037

Table 2: Normalized Knock Margin Score Values

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM	SUM
0.8	0.4779089143	0.9753428536	0.9921575762	0.9945115526	0.9999999244	0.9023849152	0.8886990141	6.23100475
0.85	0.4662966805	0.9620566879	0.9788148267	0.9811608816	0.9866307843	0.8893442627	0.8757044165	6.14000854
0.9	0.4363505839	0.9277937882	0.9444060064	0.9467316332	0.952153907	0.8557145025	0.8421934245	5.905343846

0.95	0.3909654448	0.8758662704	0.8922573371	0.8945520037	0.8999020931	0.8047465462	0.7914054687	5.549695164
1	0.3324571198	0.8089238272	0.825029796	0.8272845504	0.832541583	0.7390411233	0.7259320937	5.091210093
1.05	0.2627003496	0.7291114479	0.7448775078	0.7470846766	0.752230762	0.6607035861	0.6478712165	4.544579547
1.1	0.1832290131	0.638184124	0.6535629379	0.6557158943	0.6607355815	0.5714564957	0.5589393148	3.921823361
1.15	0.09531022764	0.5375916306	0.5525420361	0.5546350175	0.5595148711	0.4727228374	0.4605543481	3.232870968
1.2	0	0.4285421141	0.443028094	0.4450560581	0.4497843219	0.365688432	0.3538979498	2.48599697

Table 3: Normalized Torque Score Values

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM	SUM
0.8	1.069897591	0.4229407691	0.1621543822	0.5932924555	0.4904462248	0.591566216	0.2920179478	3.622315586
0.85	1.049927997	0.4592619908	0.2654552756	0.6182202257	0.5274274811	0.6119823886	0.3831382694	3.915413628
0.9	1.050282574	0.5221074288	0.3221632348	0.6757333458	0.5875101499	0.6664342875	0.4306915695	4.254922591
0.95	1.021931336	0.51766046	0.3003759169	0.6647909382	0.5823694823	0.6536039341	0.4021074555	4.142839523
1	0.9269781057	0.4855349106	0.2781155925	0.6271302749	0.5497050343	0.6150300135	0.3751262359	3.857620167
1.05	0.8185316113	0.4300461806	0.2307646784	0.5669238745	0.4937972596	0.5547353574	0.3247555121	3.419554474
1.1	0.698692769	0.3544657119	0.1618166699	0.4873374752	0.4178964369	0.475772721	0.2542560776	2.850237862
1.15	0.5689780758	0.261281261	0.08244993956	0.3907808666	0.324478976	0.3804664524	0.1746453415	2.183080913
1.2	0.4304593242	0.1632476755	-0.01507579385	0.2907022517	0.2256954858	0.2821953289	0.07804469643	1.455268969

Table 4: Normalized Performance Score Values